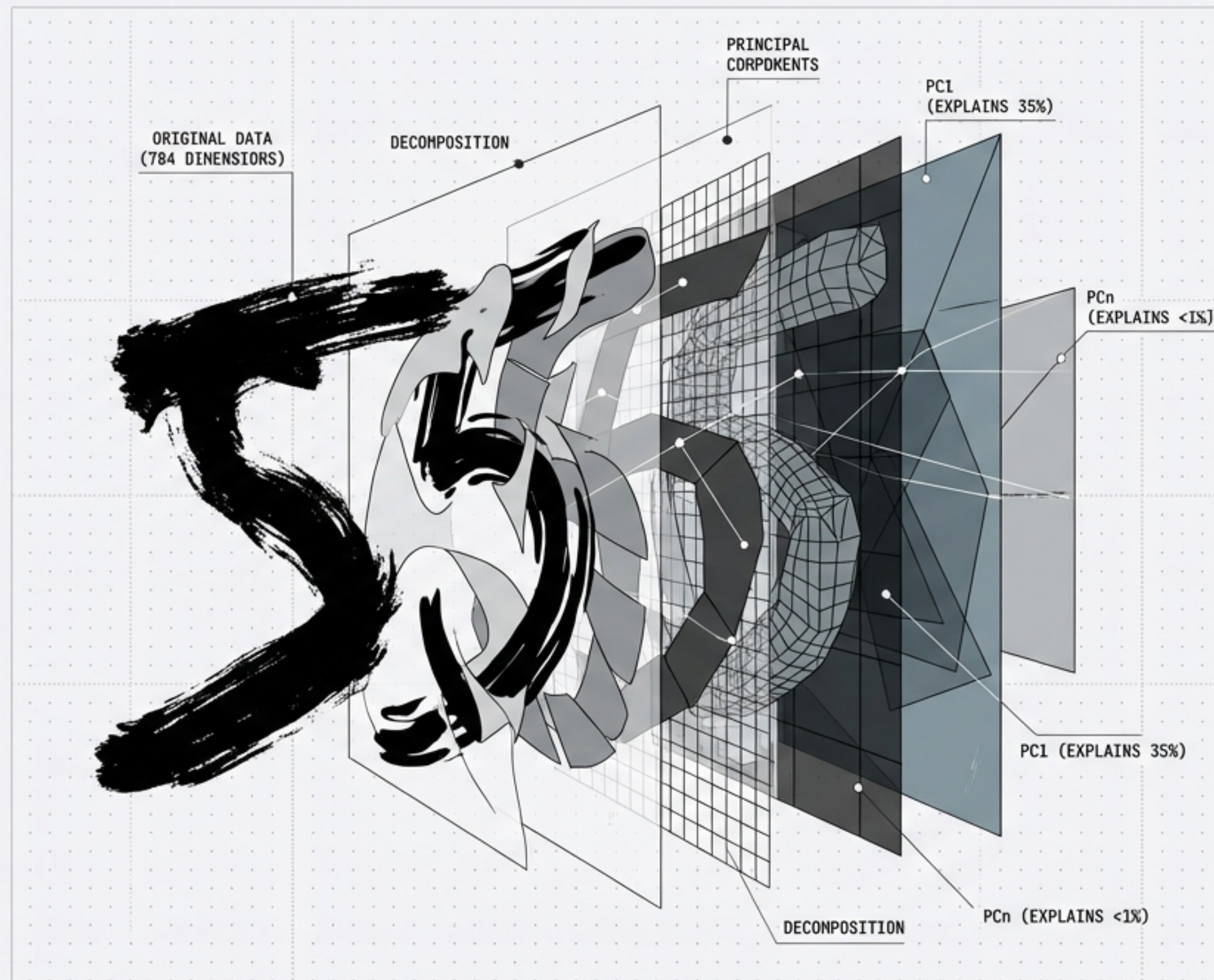


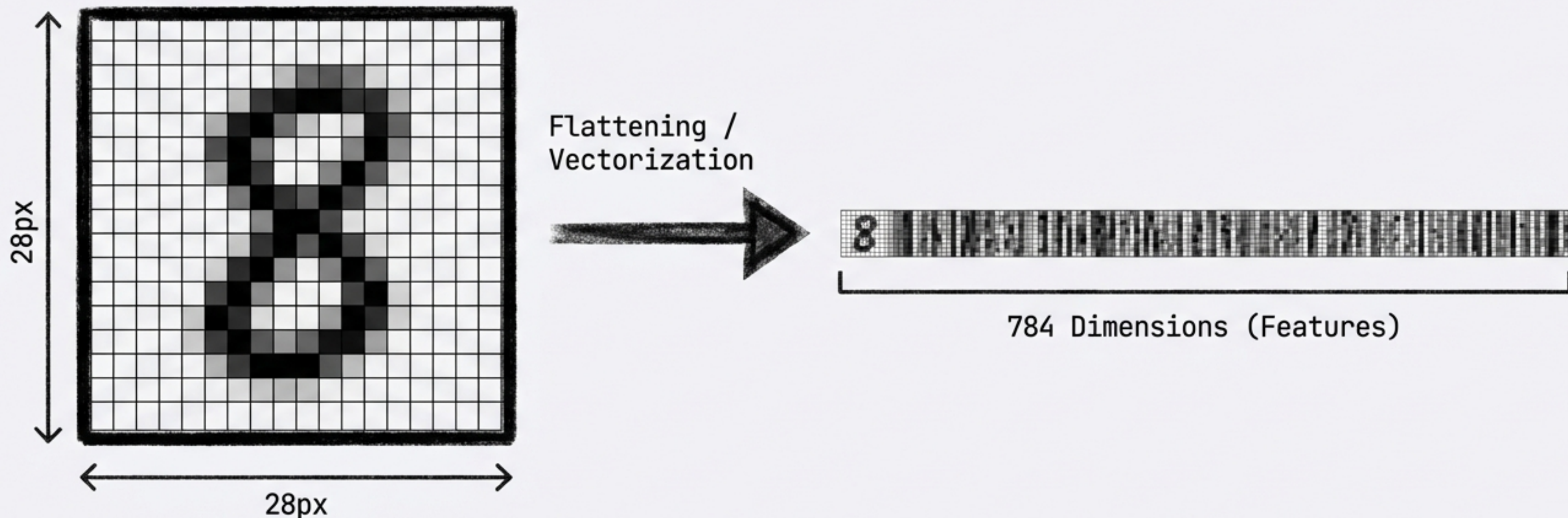
Unlocking Efficiency with Principal Component Analysis

An Empirical Case Study on the MNIST Dataset

Objective: Quantifying the trade-off between computational speed and predictive accuracy in high-dimensional machine learning models.



The Raw Material: 42,000 Images



Input:
MNIST Handwritten Digits

Volume:
42,000 Training Samples

Complexity:
784 pixels per image
= 784 independent variables

Establishing the Baseline.

Report Card

Algorithm: K-Nearest Neighbours (KNN)

Input Data: Raw 784-dimensional vectors

Distance Metric: Euclidean

The Pain

Metric

96.7% Accuracy

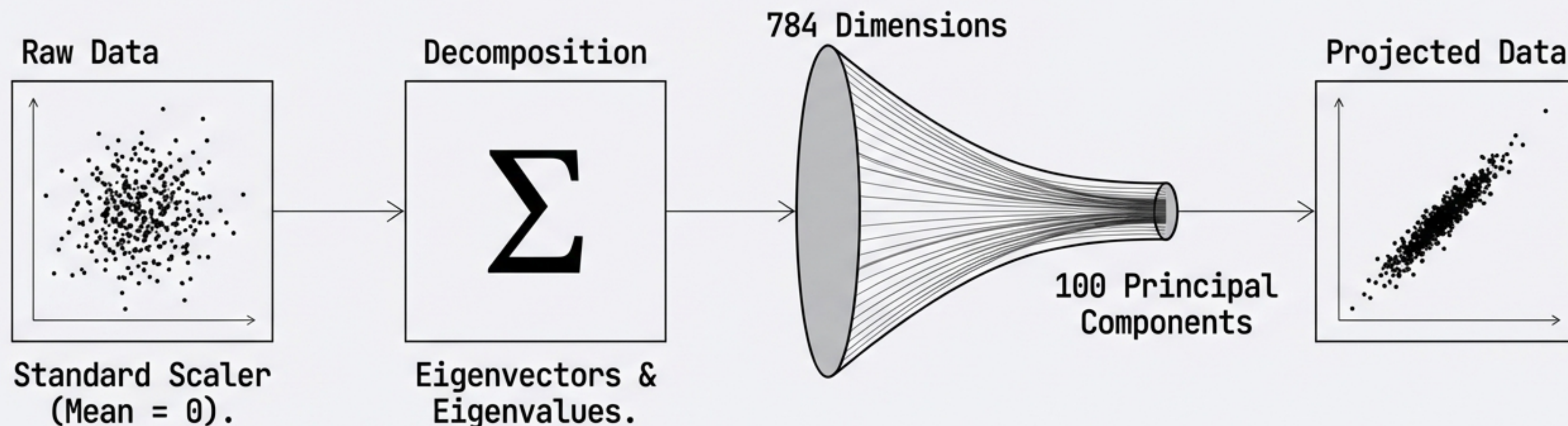


Computational Cost: High

The Curse of Dimensionality:
Calculating distances between thousands of points in 784 dimensions creates a computational bottleneck.

The Intervention: PCA.

Standardise. Decompose. Project.



Goal: Transform high-dimensional data into a new coordinate system that maximizes variance while reducing feature count.

The Trade-Off: Speed vs. Precision

Baseline (Raw Data)

Dimensions: 784

Accuracy: 96.7%

Speed: Slow

PCA Transformed

Dimensions: 100 (13% of original)

Accuracy: 95.5%

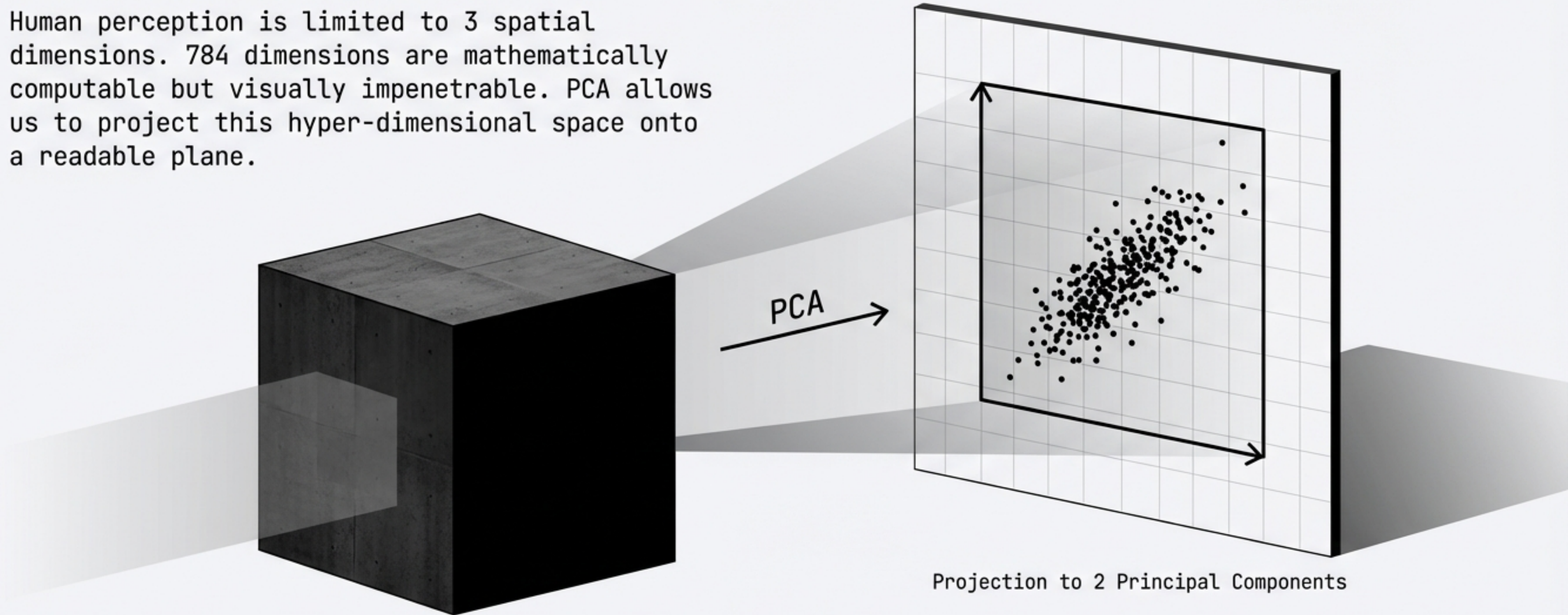
Speed: Fast



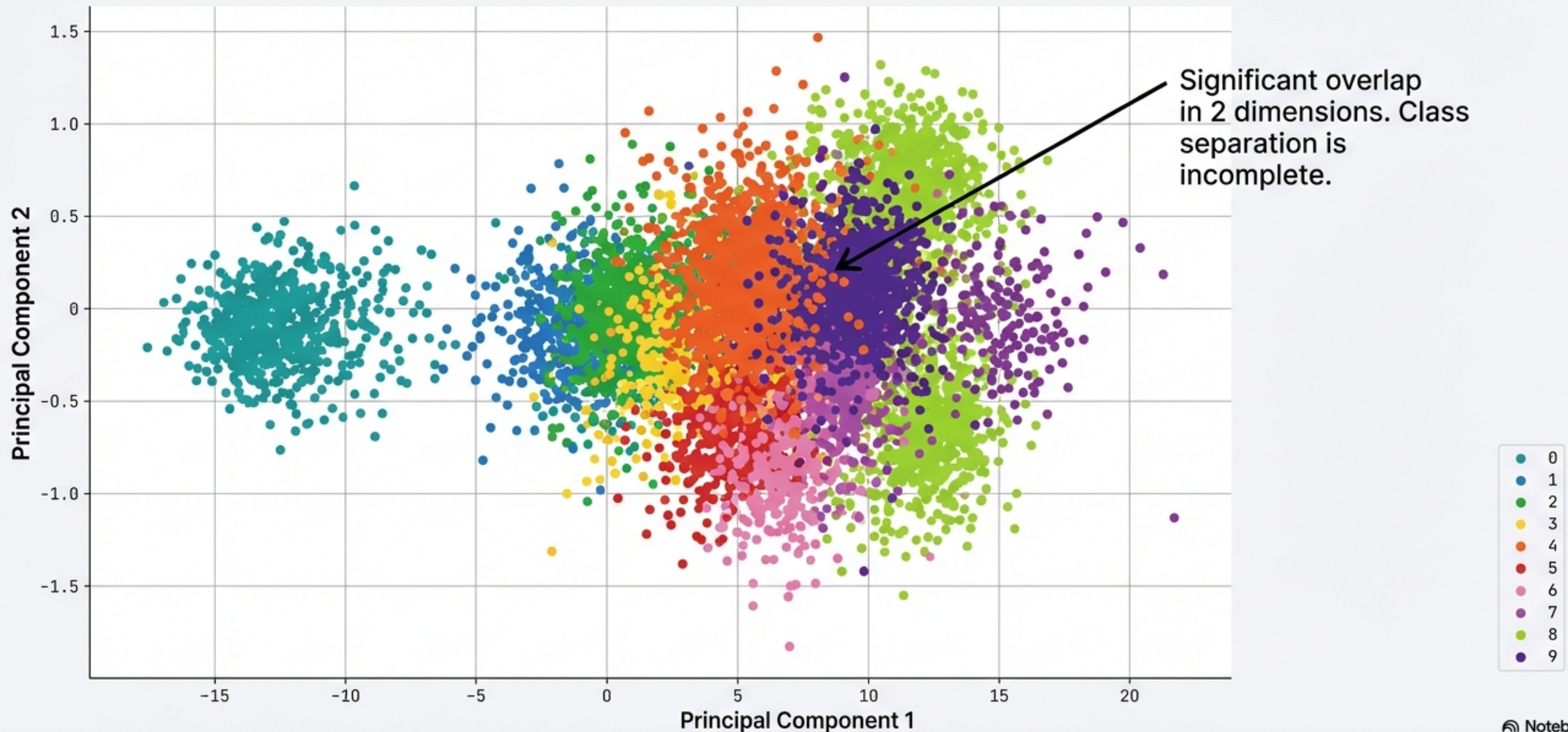
Loss: -1.2% Accuracy
Gain: +87% Compression

Visualising the Black Box

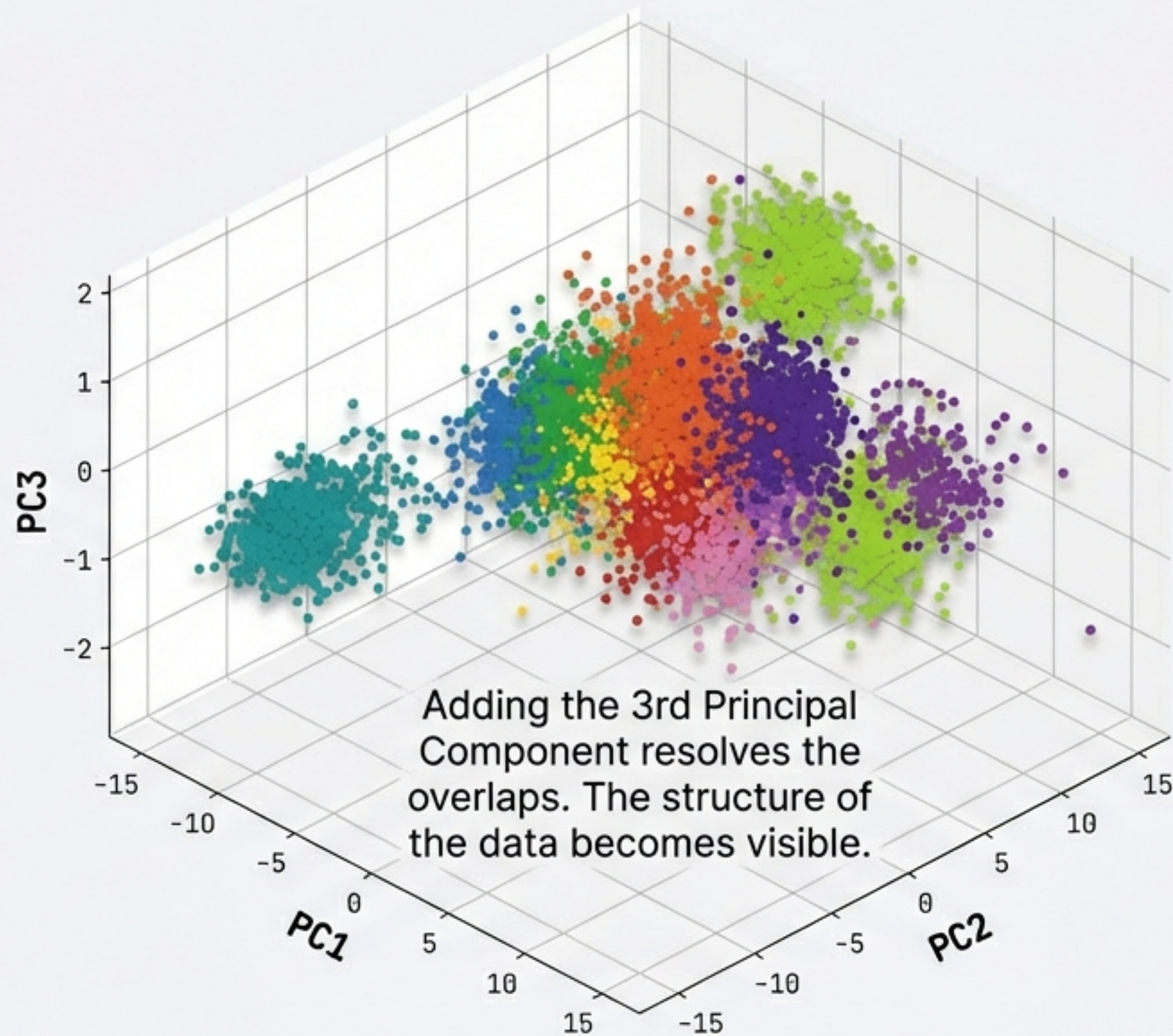
Human perception is limited to 3 spatial dimensions. 784 dimensions are mathematically computable but visually impenetrable. PCA allows us to project this hyper-dimensional space onto a readable plane.



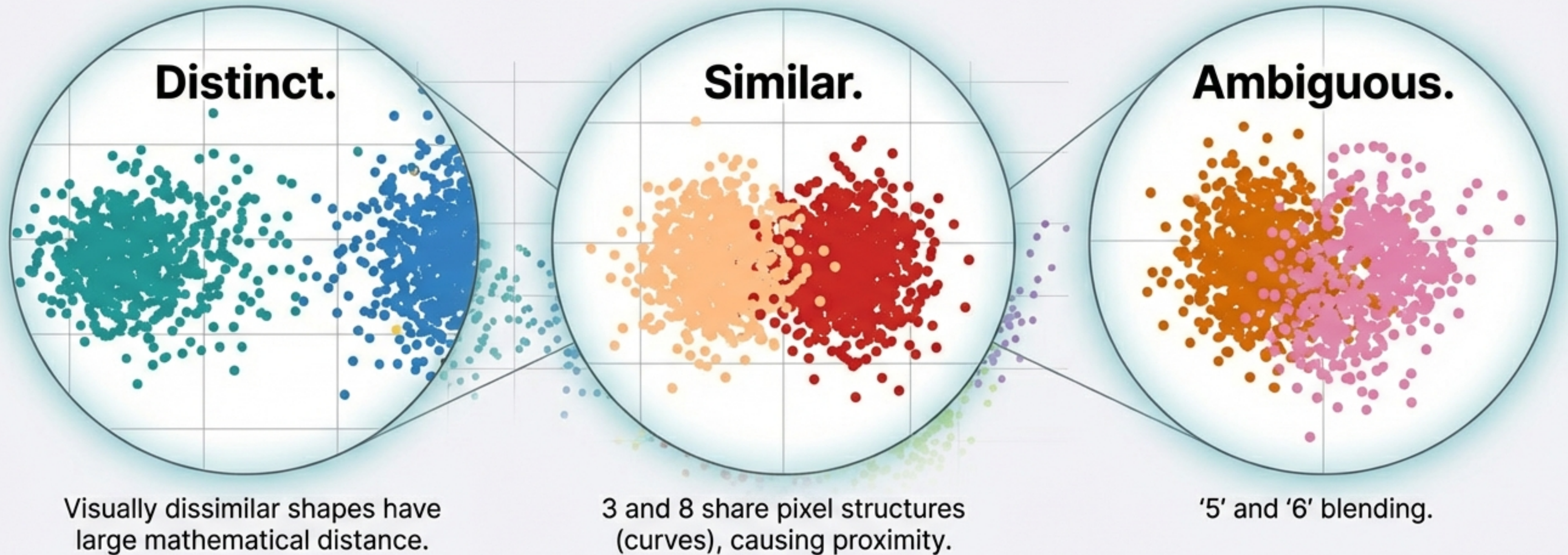
The 2D Projection.



The 3D Breakthrough.

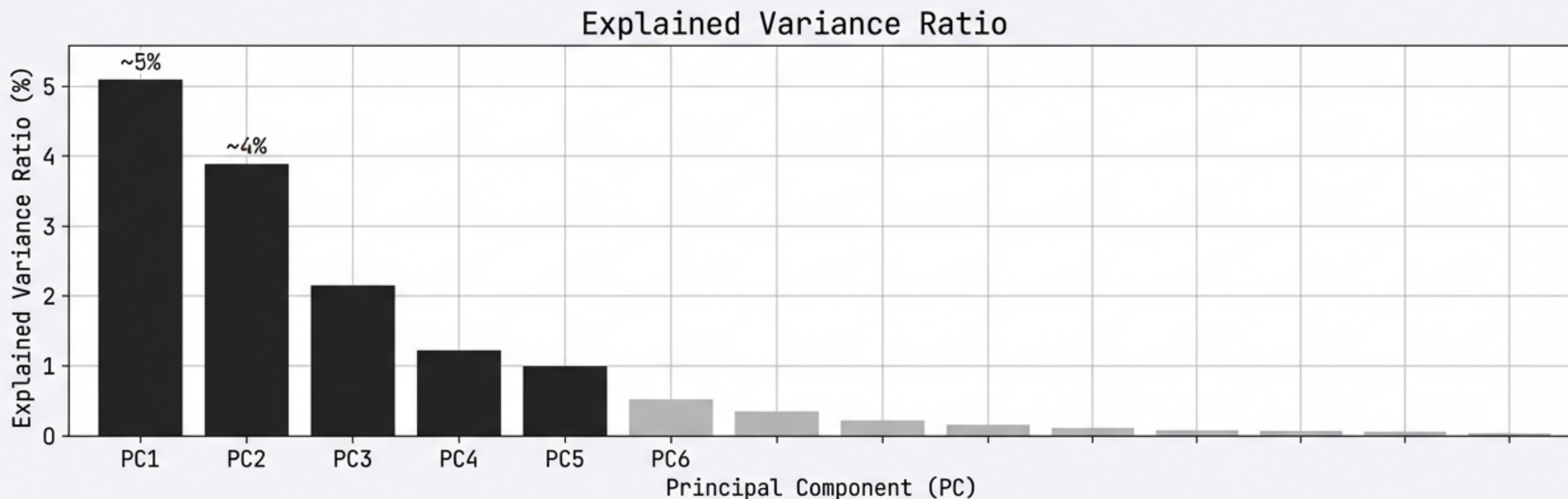


The Logic of the Clusters.



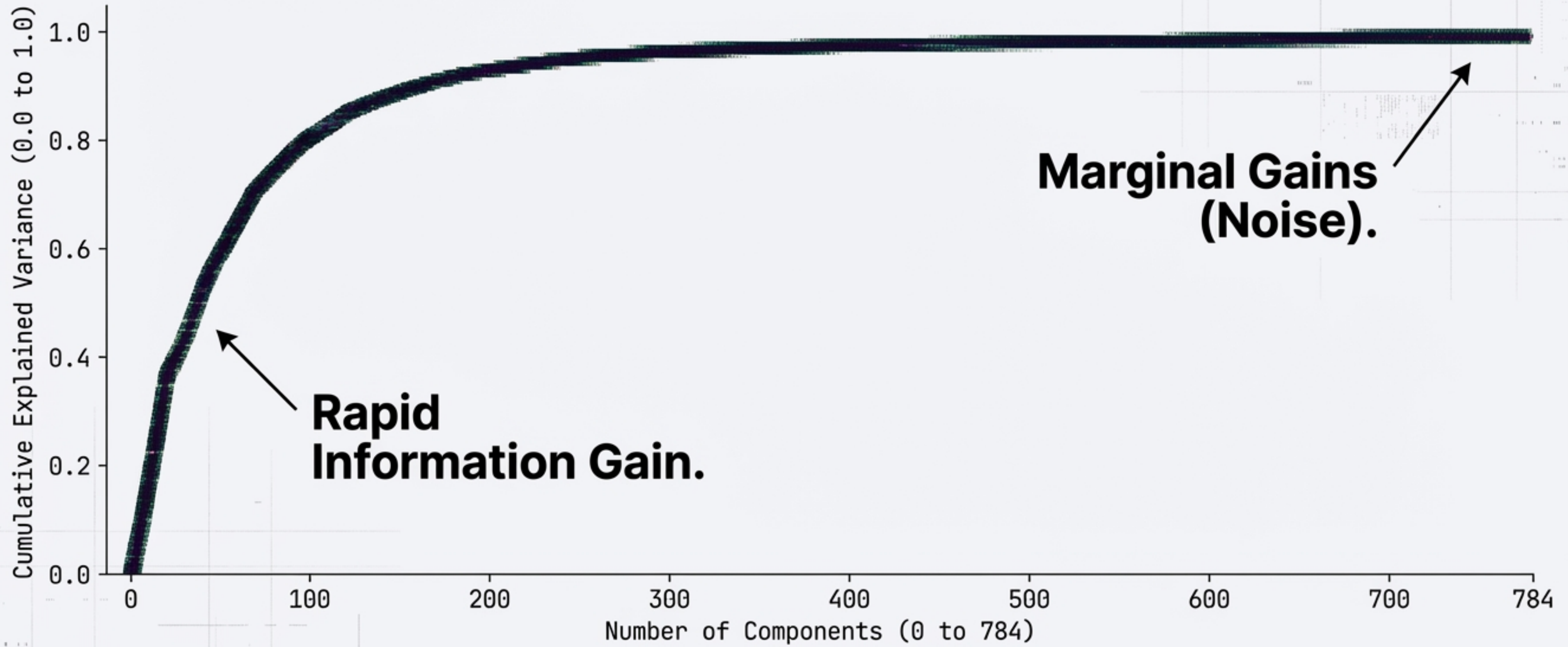
"The machine — calculates similarity the same way the human eye perceives shape."

Optimisation: Finding the Signal



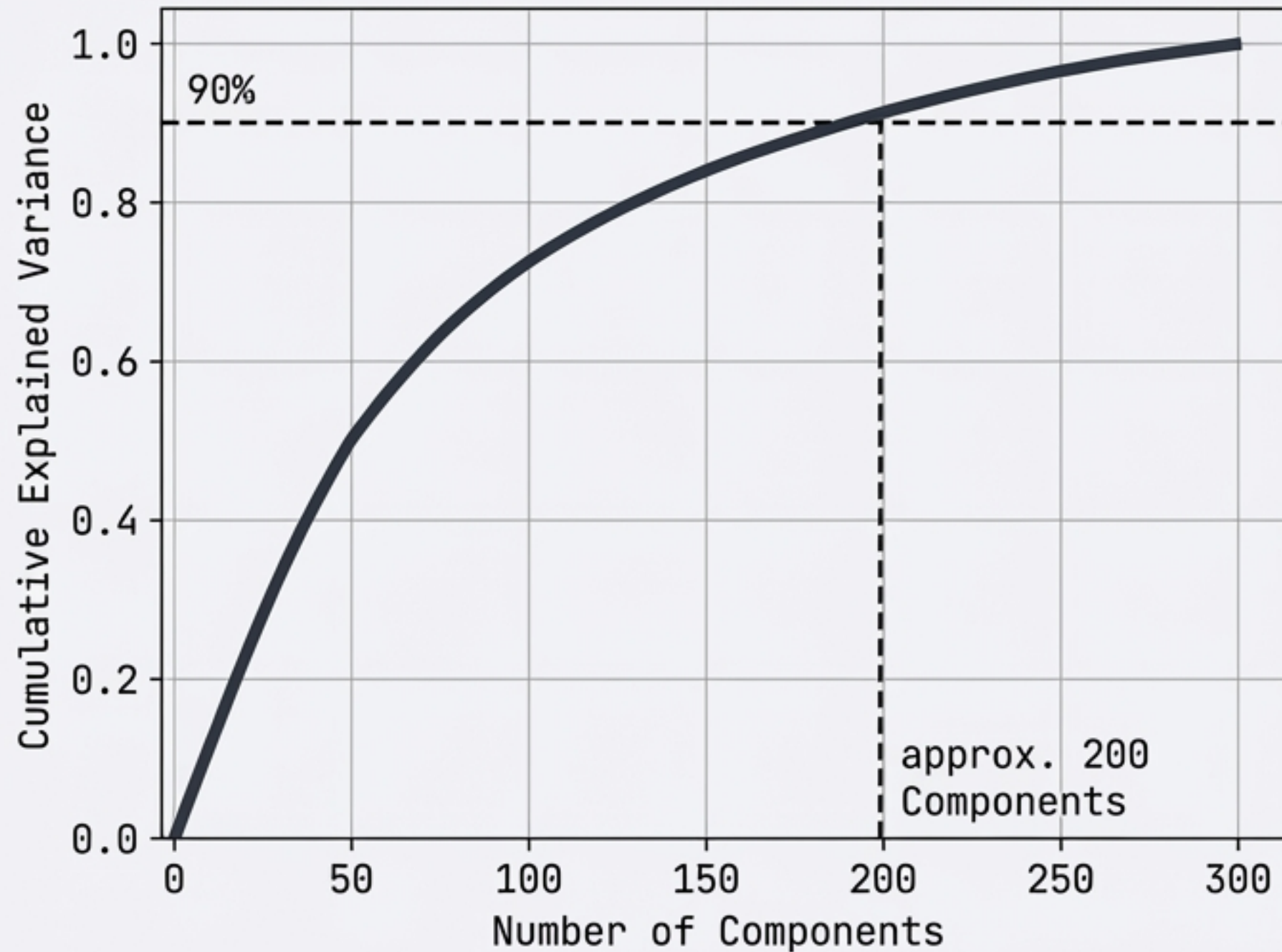
Eigenvalues (λ) measure the amount of information (variance) captured by each component. The signal is concentrated in the first few dimensions; the rest is noise.

The Law of Diminishing Returns.



The chart clearly demonstrates that you don't need all 784 components to reach a high Y-value.

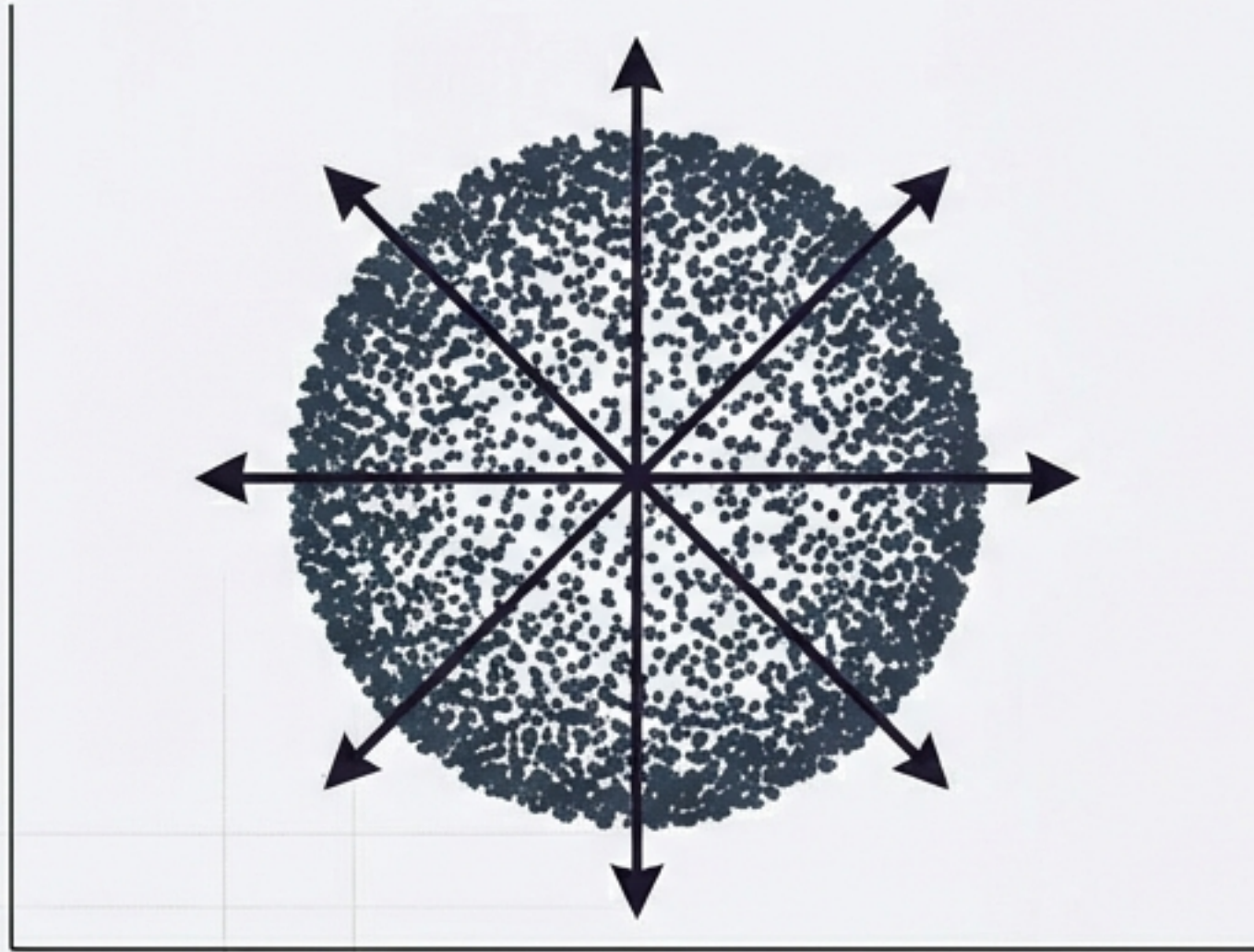
The 90% Rule.



The Sweet Spot.
We retain 90% of
the signal while
discarding 75% of
the dimensions.

Boundary Conditions: When PCA Fails.

Spherical Data.



Equal Variance: No principal direction can be found.

Non-Linear Manifolds.



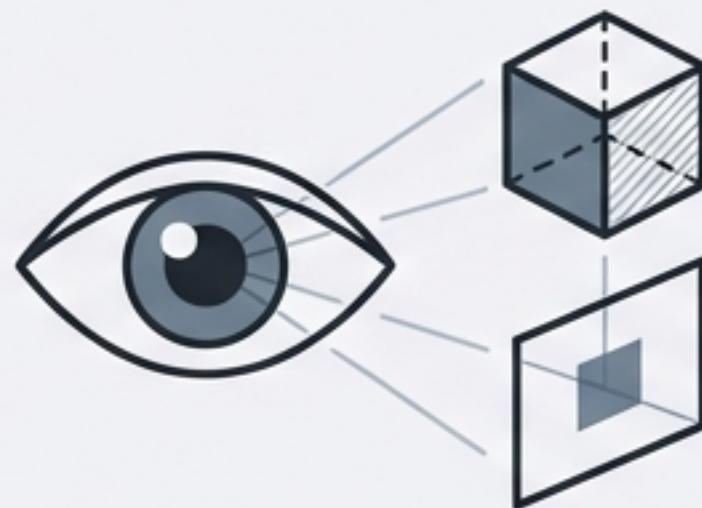
Linear Projection destroys non-linear structures (use t-SNE or UMAP instead).

Summary of Capabilities



Dimensionality Reduction

Compresses 784 features into manageable vectors, solving the Curse of Dimensionality.



Visualisation

Projects hyper-dimensional data into interpretable 2D and 3D plots.



Noise Reduction

Isolates high-variance signal and discards low-variance background noise.

The Strategic Sacrifice.

Speed + Interpretability > Absolute Precision

PCA is not about perfect reconstruction. It is about discarding the irrelevant to reveal the structural truth of the data.

Apply the 90% Variance Rule to your own datasets.